
Who This? Authorship Classification of Short Form Text Messages

Alex Oshima
aoshima@ucsd.edu

William Zhan
wzhan@ucsd.edu

Alexander Qin
aqin@ucsd.edu

Arnav Khanna
arkhanna@ucsd.edu

Yangkun Wang
yaw048@ucsd.edu

Abstract

In this paper, we attempt to use a combination of transformers, a GRU model and an LDA model to classify short text messages from different users. By analyzing the content, syntax and topics of messages, we hope to be able to successfully predict which user sent which message from a pool of 10 potential authors. After fine-tuning the models and pre-processing data in different ways, we achieved a suitable performance compared to other research in the field. This paper best helps us understand trends in instant messaging and how people communicate with each other using the internet.

1 Introduction

Over the past several years, Discord has become a popular way for gamers, friends, and teams to connect. It is an instant messaging platform that allows users to send quick messages to each other. In this study, we delve into the problem of identifying message senders in Discord channels. The task of authorship identification and classification has been around for a while and has had many gives in the field of NLP. Our research builds off this by applying these methods to short form messages. Traditional research has focused on identification of long documents such as books or news articles. Text messages are inherently much harder to identify the author of. This is due to the lack on context as well as the simplicity of the messages. i.e. "hi" "hey" etc. Our approach utilizes three machine learning models, i.e., A GRU model, the latent Dirichlet allocation (LDA), and the transformer model, and integrates them into an ensemble framework to improve the prediction accuracy.

2 Related Works

Similar research includes projects such as [1] and [3], where results have shown an ability to classify (out of a group) which user produced which body of text.

[1] focuses more on linguistic patterns and analyzing these to complete the classification task. As opposed to using what this paper refers to as "shallow" features" (word/term frequencies), this paper shows the importance of using more linguistic features, such as syntax. We plan to use both contextual and syntactical features to achieve an optimal model.

[2] This is the Pytorch Library that we will be using to conduct the processes in the paper.

3.1 Exploratory Data Analysis

Year	Number of Publications
2013	176
2014	27
2015	8
2016	4
2017	4
2018	4
2019	4
2020	4
2021	4
2022	4

[illegible]

2

3.2 Data Preprocessing

The cleaning and preprocessing of the dataset is complete. For this portion of the project, the nulls were removed from the data, and alternative characters were taken out of the text. This includes characters like newlines. Furthermore, Discord/bot generated messages were removed, as well as tokens of size < 3 . Lastly, personal information, such as passwords, were removed from the dataset. Furthermore, we only used authors who have sent over 15000 messages, and have used a random subset of those messages totalling to 15000 messages per author, meaning a total dataset of 150000 messages over 10 authors.

To transform the data, words were tokenized, only accepting string message attributes and authors.

```
bellebuddy, AHHAHAHAHAHAHA
its_ashpee,yay the piss trio kpissy ashpee mantinkle
bellebuddy,Oh my god
.hyru,
knex,.went searching for this cus i forgot what mans nickname was supposed to be
laeloria,we'll see
knex,.but for how much longer
knex,.nice
laeloria,also my wifi is working rn lol
laeloria,wow payday big game
knex,.done downloading it
laeloria,i think you have options
laeloria,kinda
knex,.is it supposed to be a stealthy game
laeloria,cant wait for all of us to play payday like a fps and with absolutely 0 stealth
laeloria,ill install it tomorrow 🙌
knex,.wow its 46 gb
knex,.im installing payday 2 rn
knex,.5 years later it finally sends
laeloria,this is my wifi
laeloria,thats my data
laeloria,yes
its_ashpee,Best wifi
knex,.wow sick
laeloria,actually nvm lemme hop on a game
laeloria,my data in my house sucks
knex,.go to starbucks to play apex
laeloria,that sounds unpleasant
laeloria,hopefully tomorrow
knex,.will your wifi ever recover
laeloria,i would play apex with jasmine but my wifi is still dying
laeloria,it was kinda poopy when we tried it
knex,.wow gamer dad
kp03,i would keep laughing out loud and my dad would occasionally join me
kp03,it was so funny
knex,.yee haw cowboy
knex,.ride horses and shoot people
knex,.red dead online looked fun
knex,.but i dont mind after lol
knex,.im eating rn
kp03,are you guys gonna finally play w her
knex,.hi jasmine
kp03,interesting
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Figure 3: Sample Messages

3.3 Model Building

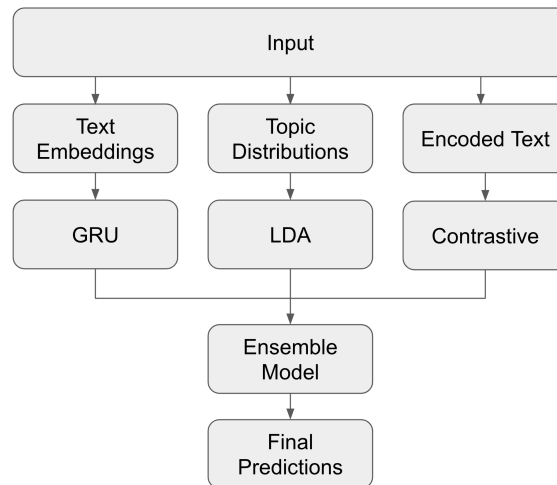


Figure 4: Model Architecture

In this subsection, we explain each component of our framework, which is depicted in 4.

Topic Extraction: After the data cleaning and the words being tokenized, each word is converted to its root (eg. runs becomes run). Then, the language is passed through a trained LDA model that generates a topic distribution for each message. In other words, a vector is created, where each element represents topic, and the value of it represents how relevant that topic is to the message. Finally, that vector is passed to a classifier and a feed-forward neural network.

Syntax Extraction: Gated Recurrent Unit (GRU) model will be trained on the language embeddings matching the tokenizer: bert-based-uncased. The goal is to utilize the Gated Recurrent Unit model to attempt to learn more of the texting syntax than from the transformer model utilized. The model essentially takes in tokenized sequences of each author and attempts to learn from those sequences to predict authorship of the message. GRU's are also proficient in analyzing different length strings and messages can vary quite drastically in string length, furthermore they are effective in filtering out noise which is often prevalent in such a dataset.

Latent Space Separation: Using a transformer trained in a contrastive manner to push messages from different authors apart in the the latent space. Hopefully this will capture any gaps that are missed by the syntax and topic extraction. The input text is first encoded with BERT's tokenizer then used to train the encoder then a neural net on top is used to classify. We train with two loops, the first of which trains the encoder. Then we freeze the encoder weights and start the second loop, which trains the neural net.

Ensemble Integration: The ensemble model processes the output from the topic extraction, syntax extraction, and latent space separation components by means of a weighted sum. We first normalize the embeddings of each model via z-score , and then multiply the output of each component by a coefficient to reflect its relative importance, where each coefficient is a predefined hyperparameter. These weighted outputs are then concatenated into a single feature vector and processed by a multi-layer perceptron (MLP). The MLP consists of several layers that perform linear transformations on the concatenated vectors, interspersed with nonlinear activation functions (e.g., ReLU) to introduce complexity and the ability to model nonlinear relationships. The final layer of the MLP outputs the predicted probabilities for each potential sender.

4 Results

Model	Test Accuracy
Ensemble model	42.75%
GRU	32.67%
Contrastive Transformer	25.67%
LDA	20.54%

Figure 5: Accuracy results of the Ensemble model, GRU, contrastive transformer and LDA.

4.1 Baseline

Our baseline model achieved an accuracy of around .29. This was based on previous research that we re-created. Our models for topic extraction, syntax extraction, and latent space separation are built upon this one.

4.2 Topic Extraction

When using only the LDA model to predict we got a test accuracy of 0.194 and loss 460. We can see that the model begins to over fit at around the 5th epoch of training.

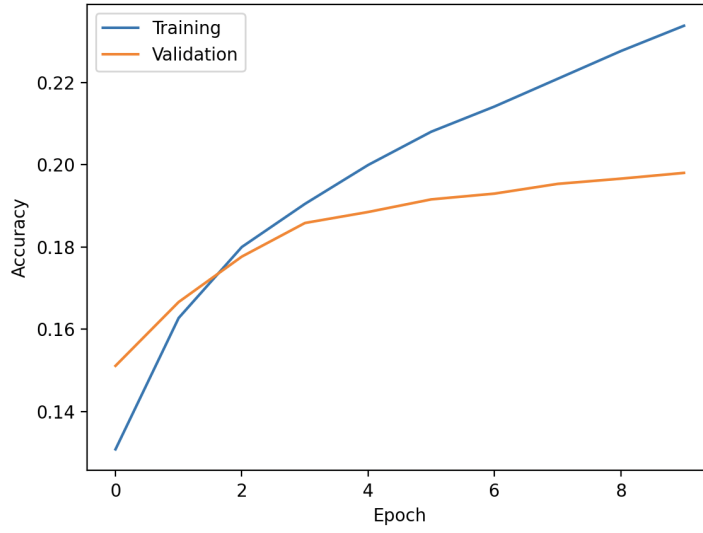


Figure 6: Plot of Training and Validation Accuracy over Epochs

4.3 Syntax Extraction

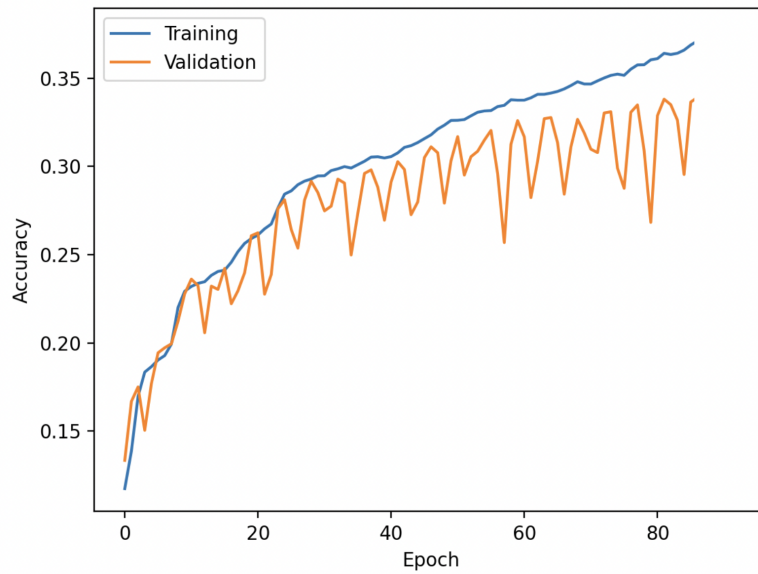


Figure 7: Plot of Training and Validation Accuracy over Epochs

For the syntax extraction process by itself, it obtained an accuracy of 32.67% with a loss of 381.42. We notice that overfitting begins to occur at epoch 90, and the peaks of the curve gradually smooth out.

4.4 Latent Space Separation

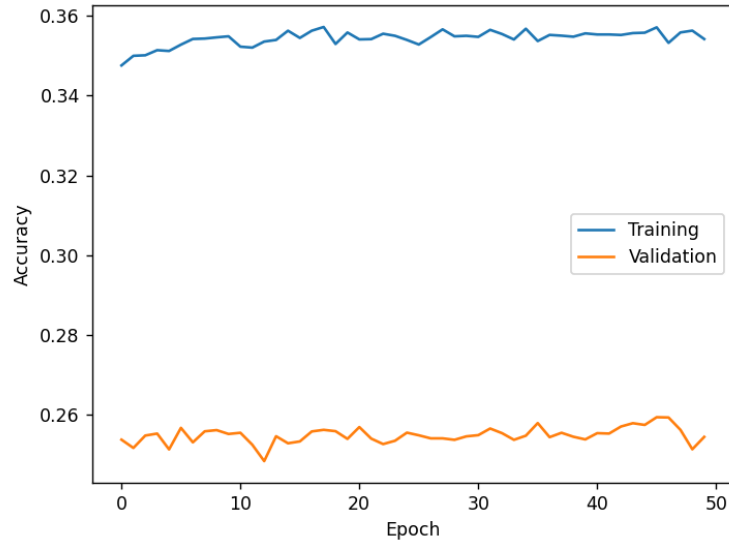


Figure 8: Training, Validation Accuracy over Epochs

The contrastive model did not see much improvement in the second training loop, reaching a validation accuracy of around .25. This is likely because the model's performance plateaued during the first training loop, which involved training the encoder as well as the neural net using supcon loss.

4.5 Ensemble

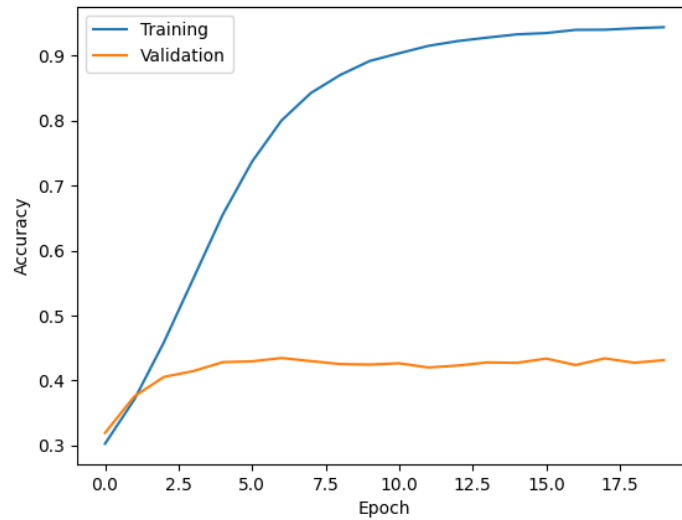


Figure 9: Accuracy of the ensemble model over epochs.

For the ensemble model, after the embeddings are concatenated, we apply a dropout layer on these embeddings for regularization. We find the optimal dropout rate to be 0.5, which achieves an accuracy

of 42.75% as in Table 5. However, as shown in Figure 9, the ensemble model severely overfits the training set, yet using a stronger regularization does not produce a better result.

5 Discussion

5.1 Baseline

Our baseline model achieved an accuracy of around .29, which when compared to our final ensemble model demonstrates an improvement of around .14. This shows that the ensemble method was effective in producing better results, although there are still potential improvements to be made that are discussed below.

5.2 Topic Extraction

Overall the inclusion of topic extraction was a very human approach to this topic. When dealing with the task of the author of a message one of the first features that people look to is the topic of the message. The LDA model does a good job at extracting the idea of a topic to a vector. Overall it was never meant to be used on its own but even when it is, it still provides over twice the accuracy of random guessing. It makes sense that over fitting occurs as when only provided with information about topics the model begins to think that certain people can only talk about certain topics. One big drawback to the LDA model is that it has to be pretrained on the corpus of all training messages so that it is able to learn the topic distribution. This pretraining took a very long time. This means that it is not very easy to scale.

5.3 Syntax Extraction

As far as Syntax Extraction utilizing the GRU network, we realize that this approach was a very general and simple approach to such a complex problem in text message recognition and syntax learning. The prevalence of short messages, as well as variation in spelling in hindsight may have caused our syntax extraction to be quite limited in its performance. Although the main idea was never to utilize syntax learning on its own, it still under performed our expected levels, simply because intuitively we do look at how people spell certain words, or phrases, or shorten certain text for recognizing the author. The graph itself illustrates how despite training to a decent degree, we still see outliers where our test or validation accuracy plummet. Although it is noted that the general trend is that the average accuracy seems to trend towards increasing with epochs meaning that there is in fact some detectable connections or features between syntax extraction and the task.

5.4 Latent Space Separation

Latent space separation using the contrastive model held mixed results. Although the model did perform better than the LDA model, and certainly better than random guessing with its accuracy of .25, the model's performance gains shrank very quickly, and after only 10 epochs it was clear the model would not improve much more. The reasons for this may be insufficient model complexity, overfitting, or just the dataset itself. To make further improvements with this model, a larger dataset and more complex architecture may be necessary, as well as further hyperparameter fine-tuning.

5.5 Ensemble

The ensemble achieved very good performance on the dataset (42.75%). However, we found that the model easily overfit on the training set. This may be due to the fact that the representations obtained by combining embeddings from different models are too expressive. However, using a higher dropout does not help the model achieve better test accuracy. Subsequently, we found that using z-score and batch norm, the generalization of the model was effectively improved. We tried using pre-trained models or training them from scratch, and found that while eventually ensemble models can achieve similar results, using pre-trained models can reduce the training epoch.

5.6 Improvements

In regards to our overall structure, we could have instead taken the optimally tuned networks from each subsection and then load it into the Ensemble Multi Layer Perceptron. This way the problem of overfitting would not have been as large of an issue. Since we trained the models in parallel essentially the LDA seemed to have been overfit, and the GRU was underfit.

Furthermore, our dataset in general could have been improved, due to the length of strings some adjustments we could have considered is simply double each string so that there would be more to learn from each data point. We could have balanced out the number of short strings and long strings per person that was being loaded in so that the model can learn both longer instances of each persons tendencies as well as shorter instances.

Although the results were less than desired, the increase in accuracy provided from the Ensemble layer does in fact illustrate that this idea can be further improved to present better results. One major pitfall was definitely the low effectiveness of our supercontrastive layer, in hindsight it may have been better to utilize more generic learning algorithms on batches of messages instead to learn more about the conversationability of the person in question.

References

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